

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Mock Interview

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Register Number	192525002
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COURSE OUTCOME COVERED

CO5: *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.*
(BL5)

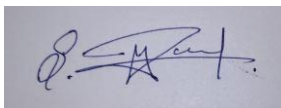
Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 25

Code of Conduct

I MAGESH.S (192525002) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student:



Assessment Tasks Mock Interview

Answer the following interview-oriented questions clearly and concisely. Support your responses with architecture-level reasoning wherever appropriate.

1. Explain the difference between HDFS and traditional file systems.
2. Why is replication important in distributed storage systems?
3. How does MapReduce divide work between map and reduce phases?
4. What role does YARN play in large-scale Hadoop processing?
5. Differentiate NAS and SAN in an enterprise data environment.
6. What is the purpose of ETL in a big data pipeline?
7. How does Apache NiFi help in data integration workflows?
8. What are the key failure points in a Hadoop cluster and how would you handle them?

9. How would you design a secure data ingestion pipeline for a large organization?

10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Mock Interview Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Knowledge	25%	Shows deep and accurate technical understanding	Good technical understanding	Basic understanding	Weak understanding
Problem Solving	20%	Responds with strong architecture-level reasoning	Reasonable reasoning	Basic reasoning	Weak reasoning
Communication	20%	Answers are confident, precise, and professional	Mostly clear communication	Moderate clarity	Weak communication
Practical Awareness	20%	Connects concepts to realworld implementation well	Some practical relevance	Limited practical relevance	No practical linkage
Confidence and Professionalism	15%	Highly confident and professional	Generally professional	Moderately professional	Low confidence and professionalism

1. Difference between HDFS and traditional file systems

HDFS (Hadoop Distributed File System) stores large files across multiple machines in a cluster. It provides fault tolerance, scalability, and high-throughput access using data replication. Traditional file systems usually store data on a single machine or storage system and are not designed for massive distributed datasets.

2. Why is replication important in distributed storage systems?

Replication means keeping multiple copies of the same data on different machines. If one machine or disk fails, another copy is available. This provides fault tolerance, high availability, and data reliability.

3. How does MapReduce divide work between map and reduce phases?

Map phase: Reads input data and converts it into intermediate key-value pairs.

Shuffle and Sort: Groups intermediate data based on keys.

Reduce phase: Processes each group and produces the final output.

Example: For counting words, Map generates (word, 1) and Reduce adds all values for each word.

4. What role does YARN play in large-scale Hadoop processing?

YARN (Yet Another Resource Negotiator) manages cluster resources and schedules applications. It decides which application gets CPU and memory resources and coordinates the execution of jobs across different nodes.

5. Differentiate NAS and SAN

NAS

SAN

Network Attached Storage

Storage Area Network

Provides file-level storage

Provides block-level storage

Accessed through a network

Usually uses a dedicated high-speed storage network

Easier to manage

More complex and costly

Common for file sharing

Common in enterprise databases and high-performance applications

6. What is the purpose of ETL in a big data pipeline?

ETL = Extract, Transform, Load.

Extract: Collect data from different sources.

Transform: Clean, filter, and convert the data.

Load: Store the processed data in a target system.

ETL helps make raw data clean, consistent, and ready for analysis.

7. How does Apache NiFi help in data integration workflows?

Apache NiFi is used to automate and manage data flows between different systems. It supports data ingestion, transformation, routing, monitoring, and tracking. Its visual interface makes it easier to design and monitor data pipelines.

8. What are the key failure points in a Hadoop cluster and how would you handle them?

Common failures include:

DataNode failure → handled using HDFS replication.

Disk failure → replace the disk and allow data recovery.

NodeManager failure → restart the service or use another node.

Network failure → provide redundant network connections.

NameNode failure → use High Availability with standby NameNode.

Regular monitoring, replication, backups, and failover mechanisms improve reliability.

9. How would you design a secure data ingestion pipeline for a large organization?

I would design it in stages:

Data Sources → Secure Ingestion → Validation → Encryption → Processing → Storage → Monitoring

Important security measures include:

TLS encryption during data transfer.

Encryption at rest.

Authentication and authorization.

Role-Based Access Control (RBAC).

Data validation and malware/error checks.

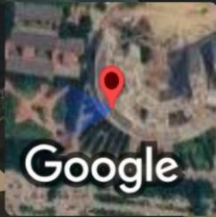
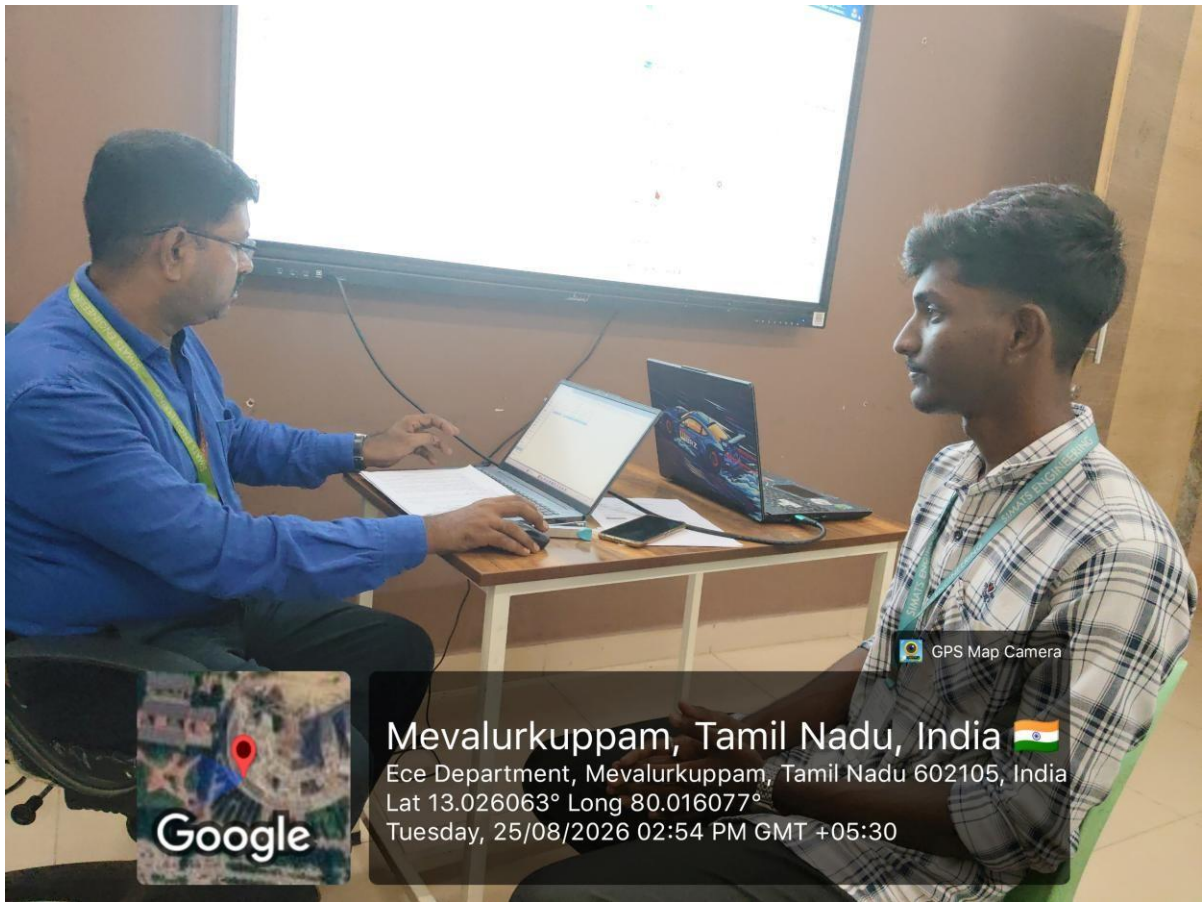
Audit logs and continuous monitoring.

This ensures that data remains confidential, accurate, and traceable.

10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

I would explain it like this:

“Think of Hadoop as a large team of workers. Instead of one computer processing a huge amount of data, Hadoop divides the work among many computers and processes it simultaneously. If the amount of data increases, we can add more computers to the team. This makes the system scalable, fault-tolerant, and cost-effective.”



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